The Moon · Teacher and Cover Plan

Learning goal: I can explain how the Moon orbits the Earth and why we see different Moon phases.

Scheme Outcome

TRACK Science scheme of work: Earth and Space

Describe the movement of the Moon relative to the Earth and explain why the Moon appears to change shape (phases) over its 28-day orbit.

Success Criteria

I can state that the Moon is a rocky sphere that orbits the Earth in about 28 days.

I can explain that the Moon does not make its own light, but reflects light from the Sun.

I can explain that Moon phases happen because we see different amounts of the sunlit half as the Moon orbits Earth.

Model Sequence (I do)

First, remember that the Sun always lights up the half of the Moon facing the Sun.

Second, as the Moon orbits the Earth over 28 days, our viewing angle of that sunlit half changes.

Third, we see the lit share grow (waxing) from New Moon to Full Moon, then shrink (waning) back to New Moon.

Why does the Moon always show the same rocky face (features and craters) toward Earth?

Made by Matt

The Moon takes about 28 days to spin once on its axis AND about 28 days to orbit the Earth.

Because its spin time matches its orbit time, the same side always faces Earth!

We Do & Hinge Check

Why is 'Earth's shadow' NOT the reason for the normal monthly phases of the Moon?

Answer: A. Normal phases happen because half of the Moon faces away from the Sun; we see different amounts of the lit half as it orbits Earth.

Safe Practical & Card Setup

Match the 8 Moon phase cards around the Earth orbit mat and compare the Space View with the Earth View.

Kit: Printed Sun-Earth-Moon Orbit Mat, 8 Space-View Moon cards, and 8 Earth-View Phase cards.

Keep desk mats flat. Never look directly at the real Sun outdoors.

You Do Routes (Supported, Independent, Aspire)

Supported: Match New Moon, First Quarter, Full Moon, and Third Quarter to their positions around the Earth and finish the sentence. (Help: Between Sun and Earth = New Moon. Opposite the Sun = Full Moon. One orbit = 28 days.)

Independent: Draw the Sun, Earth, and Moon at New Moon and Full Moon, shade the unlit half of the Moon, and explain why the Moon appears to change shape over 28 days. (Help: The Sun always lights __ of the Moon. As

the Moon orbits Earth in __ days, we see different amounts of the __ half.)

Aspire: Explain why Earth's shadow does not cause monthly Moon phases, and why we do not get a lunar eclipse at every Full Moon. (Help: Monthly phases come from viewing the Moon's own __ side. We do not get an eclipse every Full Moon because the Moon's orbit is __ by about 5 degrees.)